



SCEGGS Darlinghurst

2007

Preliminary Course
Semester 2 Examination

Centre Number

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Student Number

Mathematics

Outcomes Assessed: P2 – P8
Task Weighting: 40%

General Instructions

- Reading time – 5 minutes
- Working time – 2 hours
- Write using blue or black pen
- Attempt **all** questions and show all necessary working
- Answer all questions on the pad paper provided
- Write your Student Number at the top of each page
- **Begin each question on a new page**
- Marks will be deducted for careless or badly arranged work
- Mathematical templates, geometrical equipment and scientific calculators may be used

Total marks – 84

- Attempt Questions 1 – 7

Question	Reasoning	Comm	Calc	Marks
1	/2			/12
2	/6		/2	/12
3	/3	/1		/12
4	/3	/1	/6	/12
5	/4	/4		/12
6	/2	/6	/2	/12
7	/4	/3	/3	/12
TOTAL	/24	/15	/13	/84

Course Number

Student Number

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SCEGGS Bathingurst

2007

Preliminary Course
Semester 2 Examination

Mathematics

Outcome Assessed: P2 – P6
Task Weighting: 40%

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Total marks – 82

• Answer Questions 1–7

Question Number	Question	Mark
1	12	10
2	16	12
3	17	12
4	17	10
5	14	10
6	15	10
7	16	10
Total	82	82

General Instructions

- Reading time – 5 minutes
- Working time – 2 hours
- Write your student number at the top of each page
- Answer all questions on the back paper
- Answer all questions and show all necessary working
- Write your student number at the top of each page
- Begin each question on a new page
- Marks will be deducted for careless or badly presented work
- Misplacement, omission, rounding, approximation and essential calculation may be used

Question 1 (12 marks)

(a) Find the value of $\sqrt{\frac{15.26 + 31.98}{(4.75)^3 \times 3.5}}$ correct to 3 significant figures. 2

(b) Express $0.\dot{6}4\dot{3}$ in simplest fraction form 2

(c) Find the values of a and b if: 2

$$\frac{\sqrt{5} + 1}{\sqrt{5} - 1} = a + b\sqrt{5}$$

(d) Simplify $\frac{x^2 - 144}{x^3 + 8} \div \frac{x + 12}{x + 2}$. 2

(e) Find the exact value of $\cos 210^\circ$. 2

(f) Solve $|x - 2| < 3$ 2

- Start a new page
-

Marks

Question 2 (12 marks)

- (a) (i) Find the gradient of the curve $y = 6 + 4x - 2x^2$ at the point $P(1, 8)$. 2

- (ii) What is the maximum value of the parabola $y = 6 + 4x - 2x^2$. 1

- (b) A function is defined as:

$$f(x) = \begin{cases} 4 - x & \text{for } x < -2 \\ 5 & \text{for } -2 \leq x < 0 \\ x^3 + 1 & \text{for } x \geq 0 \end{cases}$$

Calculate the value of:

- (i) $f(-10)$ 1

- (ii) $f(-2)$ 1

- (iii) $f(m^2)$ 1

- (c) If $\cos \theta = -\frac{2}{3}$ and $\tan \theta > 0$ find the exact value of $\operatorname{cosec} \theta$. 3

- (d) Find the values of k for which the equation $x^2 - (k + 3)x + 4k = 0$ has:

- (i) equal roots. 2

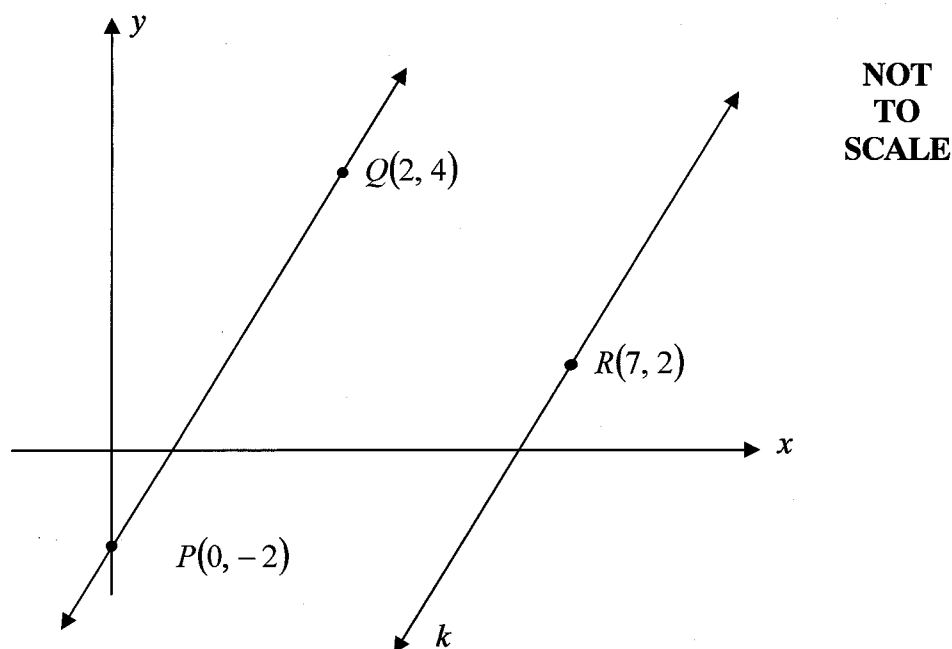
- (ii) roots which are equal in magnitude but opposite in sign. 1

- Start a new page

Marks

Question 3 (12 Marks)

(a)



In the diagram $P(0, -2)$, $Q(2, 4)$ and $R(7, 2)$ are points on the number plane.

Copy or trace the diagram onto your worksheet.

- | | | |
|-------|--|---|
| (i) | Find the length of PQ . | 1 |
| (ii) | Find the equation of the line through PQ . | 1 |
| (iii) | Find the equation of the line k passing through R parallel to PQ . | 2 |
| (iv) | The point S lies on k such that QR is parallel to PS .
What type of quadrilateral is $PQRS$. (Give reasons.) | 1 |
| (v) | Find the co-ordinates of S . | 2 |
| (vi) | Find the perpendicular distance from R to PQ . | 2 |
| (vii) | Find the area of $PQRS$. | 1 |
- (b) Solve $\sqrt{3} \tan \theta + 1 = 0$ for $0^\circ \leq \theta \leq 360^\circ$. 2

- Start a new page
-

Marks

Question 4 (12 Marks)

(a) Differentiate the following:

(i) $y = 5x^3 - \frac{1}{3x^2}$ 2

(ii) $y = x(4x + 3)^5$ 2

(iii) $y = \frac{6x - 1}{4 - 2x}$ 2

(b) Holly and Ellen go bushwalking. They leave their campsite and set out on different directions. Holly walks due East for 2.5km while Ellen walks 1.7km on a bearing of $250^\circ T$

(i) Draw a diagram showing the above information. 1

(ii) Calculate the distance between Holly and Ellen. 2

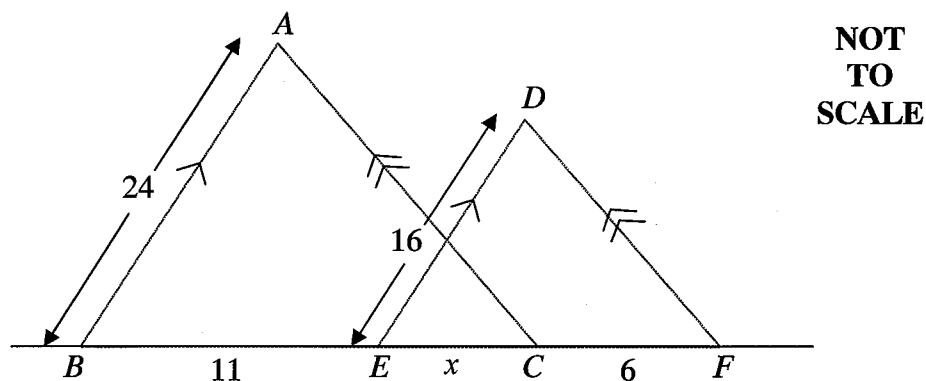
(iii) Calculate the bearing of Holly from Ellen. 3

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Marks

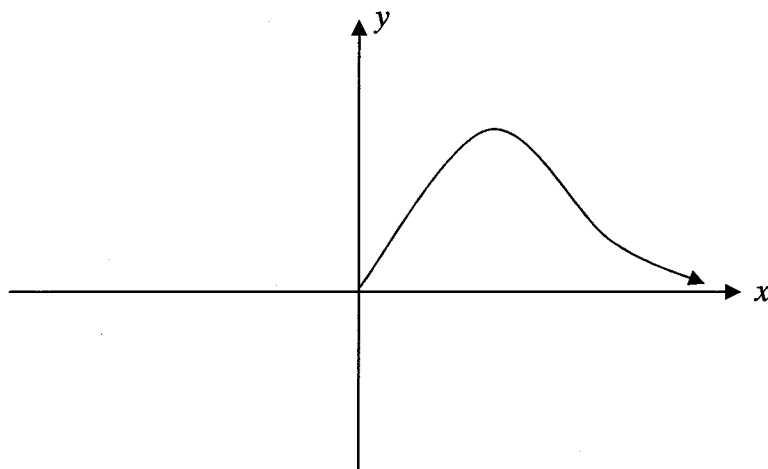
Question 5 (12 marks)

(a)



In the diagram $AB \parallel DE$ and $AC \parallel DF$. $AB = 24$, $DE = 16$, $BE = 11$ and $CF = 6$.

- | | | |
|------|--|----------|
| (i) | Prove $\triangle ABC \sim \triangle DEF$ | 3 |
| (ii) | Find EC . | 2 |
- (b) The diagram below shows part of the graph of $y = f(x)$.
You are told that $f(x)$ is an odd function. **1**



Copy the diagram and complete the graph of the function.

Question 5 continues on the next page

Question 5 (continued)

(c) If α and β are the roots of the equation $x^2 - 2x + 4 = 0$ find the value of:

- | | | |
|-------|--|----------|
| (i) | $\alpha + \beta$ | 1 |
| (ii) | $\alpha\beta$ | 1 |
| (iii) | $\alpha^2 + \beta^2$ | 2 |
| (iv) | $\left(\alpha - \frac{1}{\beta}\right)\left(\beta + \frac{1}{\alpha}\right)$ | 2 |

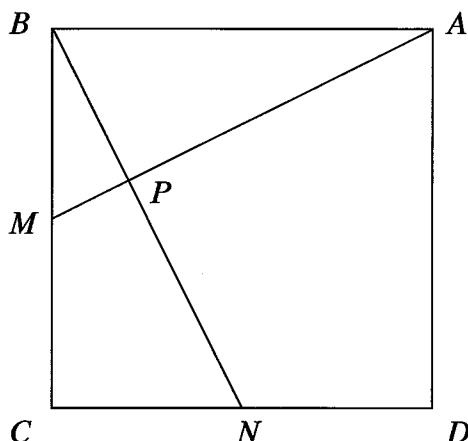
End of Question 5

- Start a new page

Marks

Question 6 (12 marks)

- (a) $ABCD$ is a square. M and N are the midpoints of BC and CD .
 BN and AM intersect at P .

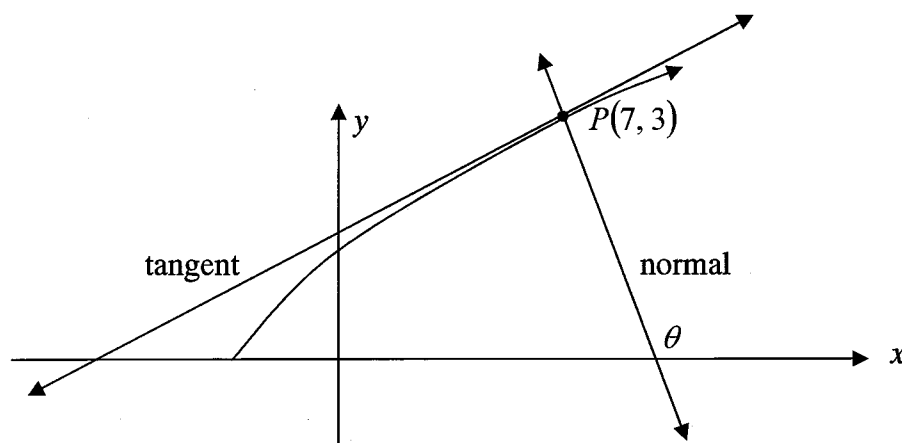


Copy or trace the diagram onto your paper showing all given information.

- (i) Prove $\triangle ABM \equiv \triangle BCN$. 3
- (ii) Prove that AM and BN are perpendicular. 2
- (b) Given the function $y = x^3 - x^2$. Find the value(s) of x for which $\frac{dy}{dx} = 8$. 2
- (c) (i) Graph $y = |x|$ and $y = 2x + 1$ on the same number plane labelling all important features. 2
- (ii) Using your graph, or otherwise explain why the equation $2x + 1 = |x|$ has only one solution. 1
- (iii) Solve $2x + 1 = |x|$. 2

Question 7 (12 marks)

- (a) The diagram below is the graph of $y = \sqrt{x+2}$. The tangent and the normal at the point $P(7, 3)$ has also been drawn



- (i) Find the equation of the normal at P . 3
- (ii) Calculate θ , the angle the normal makes with the position x -axis. 1
- (b) (i) On the same number plane graph $y = \sqrt{16-x^2}$ and $2x + y = 4$. 2
- (ii) Shade the region on your diagram that satisfies the following: 1
- $$y \leq \sqrt{16-x^2}, 2x + y \geq 4 \text{ and } y \geq 0$$
- (iii) Calculate the area of the shaded region. 1
- (c) (i) Show that $\tan \theta + \cot \theta = \frac{1}{\sin \theta \cos \theta}$. 1
- (ii) Hence, or otherwise, solve 3

$$\frac{1 + \cot \theta}{\operatorname{cosec} \theta} - \frac{\sec \theta}{\tan \theta + \cot \theta} = -1 \quad 0^\circ \leq \theta \leq 360^\circ$$

End of paper

Y11 MATHEMATICS SEMESTER 2 EXAMINATION - SOLUTIONS

Q1 a) 0.355 ✓ (1 for correct rounding)

b) Let $x = 0.64343\dots$

Reason - 2

$10x = 6.4343\dots$ ① ✓

$1000x = 643.4343\dots$ ②

② - ①

$990x = 637$

$x = \frac{637}{990}$ ✓

c) $\frac{\sqrt{5}+1}{\sqrt{5}-1} \times \frac{\sqrt{5}+1}{\sqrt{5}+1}$ ✓

$= \frac{5 + \sqrt{5} + \sqrt{5} + 1}{5 - 1}$

$= \frac{6 + 2\sqrt{5}}{4}$

$= \frac{3 + \sqrt{5}}{2}$

$\therefore a = \frac{3}{2} \quad b = \frac{1}{2}$ ✓

d) $\frac{x^2 - 144}{x^2 + 8} \div \frac{x+12}{x+2}$

$= \frac{(x+12)(x-12)}{(x+2)(x^2+2x+4)} \times \frac{x+2}{x+12}$ ✓

$= \frac{x-12}{x^2+2x+4}$ ✓

e) $\cos 210^\circ = \cos(180^\circ + 30^\circ)$ ✓

$= -\cos 30^\circ$

$= -\frac{\sqrt{3}}{2}$ ✓

f) $|x-2| < 3$

$-3 < x-2 < 3$ ✓

$-1 < x < 5$ ✓

- done well - some students need to revise significant figures

- most students knew the process but some got confused as to what quantity to multiply the x by.

- most students knew to rationalise the denominator but some were confused with expressing the final answer as $a = \frac{3}{2} \quad b = \frac{1}{2}$

- done fairly well

- students need to learn to express angles as either $180 - \theta$ or $180 + \theta$ or $360 - \theta$ } this area needs review

- a number of students treated this as two separate inequalities you should state it as in the solutions

Q2 a) i) $y = 6 + 4x - 2x^2$
 $\frac{dy}{dx} = 4 - 4x$ ✓
 when $x=1$

$\frac{dy}{dx} = 4 - 4 \times 1$
 $= 0$ ✓

Calculus - 2

$\therefore$ gradient of the curve at $P(1, 8)$ is 0.

ii) since the gradient of the curve at P is 0
 then P is the vertex.

$\therefore$ max value = 8 ✓

Reason - 1

b) i) $f(-10) = 4 - (-10)$
 $= 14$ ✓

ii) $f(-2) = 5$ ✓

iii) $f(m^2) = (m^2)^3 + 1$
 $= m^6 + 1$ ✓

Reason - 1

c) $\cos \theta = -\frac{2}{3}$

$\cos \theta < 0$ $\tan \theta > 0$

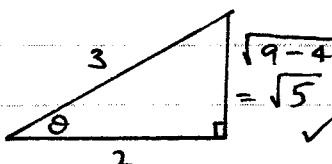
$\therefore \theta$ is in the 3rd quadrant

$\therefore \sin \theta < 0$ ✓

$\sin \theta = -\frac{\sqrt{5}}{3}$

$\therefore \csc \theta = -\frac{3}{\sqrt{5}}$ ✓

Reason - 3



Some students didn't realise the y had to differentiate

Most students wrote the vertex max value is the y value only.

Well done (i) and (ii)

iii) Students identified the correct function but failed to sub m^2 in correctly
Index laws

On a whole well done
 Students lost 1 mark for not identifying correct as neg.

d) i) $\Delta = 0$ for equal roots

$\Delta = b^2 - 4ac$

$= (-(k+3))^2 - 4 \times 1 \times 4k$

$= k^2 + 6k + 9 - 16k$

$= k^2 - 10k + 9$ ✓

$\therefore k^2 - 10k + 9 = 0$

$(k-9)(k-1) = 0$

$k=1$ or $k=9$ ✓

A lot of students incorrectly substituted c , getting an incorrect quadratic.
 $\Delta = 0$.

let $S(x, y)$ and $Q(2, 4)$

$$\therefore \frac{x+2}{2} = \frac{7}{2} \quad \frac{y+4}{2} = 0$$

$$x+2 = 7 \quad y+4 = 0$$

$$x = 5$$

$$y = -4$$

$$\therefore S(5, -4) \quad \checkmark$$

Reason - 2

vi) PQ: $3x - y - 2 = 0$ R(7, 2) $\checkmark$

$$d = \frac{|3 \times 7 - 2 - 2|}{\sqrt{3^2 + (-1)^2}} \quad \checkmark$$

$$= \frac{17}{\sqrt{10}} \text{ units} \quad \checkmark$$

Learn the correct formula.

$$\left| \frac{ax_1 + by_1 + c}{\sqrt{a^2 + b^2}} \right|$$

vii) $A = bh$

$$= 2\sqrt{10} \times \frac{17}{\sqrt{10}}$$

$$= 34 \text{ units}^2 \quad \checkmark$$

Reason - 1

Make sure you know area formula !!!

Parallelograms are easy.
 $A = bh$.

b) $\sqrt{3} \tan \theta + 1 = 0$

$$\sqrt{3} \tan \theta = -1$$

$$\tan \theta = -\frac{1}{\sqrt{3}}$$

θ lies in the 2nd & 4th quadrants

$$\therefore \theta = 30^\circ$$

$$\therefore \theta = 180^\circ - 30^\circ \quad \theta = 360^\circ - 30^\circ$$

$$= 150^\circ$$

$$= 330^\circ \quad \checkmark$$

one mark for correct rearrangement and correct angle using either 30° or 30° .

second mark angles found in quadrants 2 & 4 since $\tan \theta < 0$.

Q4 a) i) $y = 5x^3 - \frac{1}{3}x^{-2}$ $\checkmark$

$$\frac{dy}{dx} = 15x^2 + \frac{2}{3}x^{-3} \quad \checkmark$$

$$= 15x^2 + \frac{2}{3x^3}$$

- expressing $\frac{1}{3x^2}$ as $\frac{1}{3}x^{-2}$ was a problem for many students

ii) $y = x(4x+3)^5$

$$\frac{dy}{dx} = v \frac{dy}{dv} + u \frac{dv}{dx}$$

$$= (4x+3)^5 \times 1 + x \times 5(4x+3)^4 \times 4$$

- done poorly because most students didn't recognize it was a product rule question

$$x(4x+3)^5$$

$$ii) \quad x + (-x) = -\frac{b}{a}$$

$$0 = \frac{k+3}{1}$$

$$k+3=0$$

$$k = -3 \quad \checkmark$$

Reason - 1

Some students did not recognise equal in mag + opp in sign $= -\frac{b}{a}$

$$\begin{aligned} Q3 \quad a) i) \quad PQ &= \sqrt{(2-0)^2 + (4-(-2))^2} \\ &= \sqrt{2^2 + 6^2} \\ &= \sqrt{40} \\ &= 2\sqrt{10} \text{ units} \quad \checkmark \end{aligned}$$

This part was well done. Leave your answer exact unless otherwise requested.

$$\begin{aligned} ii) \quad M_{PQ} &= \frac{4-(-2)}{2-0} \\ &= \frac{6}{2} \\ &= 3 \end{aligned}$$

$$\text{using } y - y_1 = m(x - x_1)$$

$$y - (-2) = 3(x - 0)$$

$$y + 2 = 3x$$

$$3x - y - 2 = 0 \quad \checkmark$$

make sure you know your formulas.

$$\text{gradient} \quad m = \frac{y_2 - y_1}{x_2 - x_1}$$

$$\text{Equation of a line} \quad y - y_1 = m(x - x_1)$$

$$iii) \quad k \parallel PQ$$

$$\therefore M_k = 3 \quad \checkmark$$

$$y - 2 = 3(x - 7)$$

$$y - 2 = 3x - 21$$

$$3x - y - 19 = 0 \quad \checkmark$$

parallel lines have equal gradients.

iv) PQRS is a parallelogram as both pairs of opposite sides are parallel $\checkmark$

Commentation - 1

Note You do not know the sides lengths unless you have found them all.

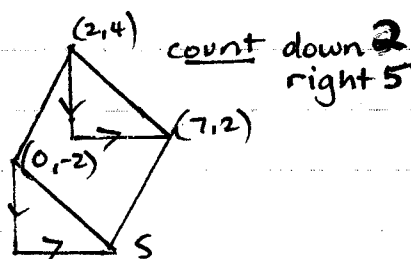
v) PR and QS has a common midpoint as the diagonals of a parallelogram bisect each other

$$\therefore \text{midpt PR} = \left(\frac{0+7}{2}, \frac{-2+2}{2} \right)$$

$$= \left(\frac{7}{2}, 0 \right) \quad \checkmark$$

$$\therefore \text{midpt QS} = \left(\frac{7}{2}, 0 \right)$$

You can just do part v) by counting.



$$\therefore S(5, -4)$$

$$= (4x+3)^4 [4x+3 + 20x]$$

$$= (4x+3)^4 (24x+3)$$

iii) $y = \frac{6x-1}{4-2x} \quad \frac{u}{v}$

$$\frac{dy}{dx} = \frac{v \frac{du}{dx} - u \frac{dv}{dx}}{v^2}$$

$$= \frac{(4-2x) \times 6 - (6x-1) \times -2}{(4-2x)^2} \quad \checkmark$$

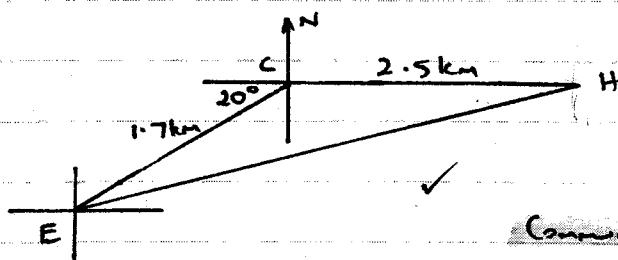
$$= \frac{24 - 12x + 12x - 2}{(4-2x)^2}$$

$$= \frac{22}{(4-2x)^2} \quad \checkmark$$

Calculus - 6

- done very well. A majority of students knew the rule and executed the process well.

b) i)



Communication - 1

- diagrams done well

ii) $\angle ECH = 250^\circ - 90^\circ$
 $= 160^\circ$

$$\therefore EH^2 = 2.5^2 + 1.7^2 - 2 \times 2.5 \times 1.7 \times \cos 160^\circ \quad \checkmark$$

$$= 17.127 \dots$$

$$EH = \sqrt{17.127 \dots}$$

$$= 4.1 \text{ (to 1 dec. pl.)} \quad \checkmark$$

$\therefore$ the distance between Holly and Ellen is 4.1 km.

- this was done well the only problem was finding the angle.

try to remember to use units.

iii) Let $\angle CEH = \theta$

$$\frac{\sin \theta}{2.5} = \frac{\sin 160}{4.1} \quad \checkmark$$

$$\sin \theta = \frac{2.5 \times \sin 160}{4.1}$$

- it is important here not to use the rounded off value of 4.1

- use the memory function on your calculator to use the most accurate answer for EH

$$= 0.2066...$$

$$\therefore \theta = \sin^{-1}(0.2066...)$$

$$= 12^\circ \text{ (to nearest degree)} \quad \checkmark$$

$$\therefore \text{bearing} = 70 + 12 \\ = 82^\circ$$

$\therefore$ the bearing of Abby from Ellen is 082°T ✓

Reasoning - 3

Q5 a) i) In Δ 's ABC and DEF

$\angle ABC = \angle DEF$ (corresponding angles in ✓
parallel lines $AB \parallel DE$)

$\angle ACB = \angle DFE$ (corresponding angles in ✓
parallel lines $AC \parallel DF$)

$\therefore \Delta ABC \parallel \Delta DEF$ (equiangular). ✓

Communication - 3

ii) $\frac{AB}{DE} = \frac{BC}{EF}$ (corresponding sides in similar,
triangles are in the same ratio)

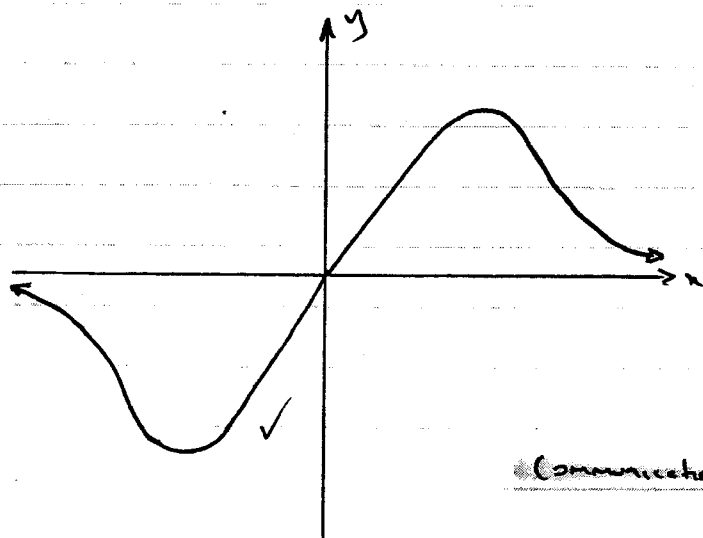
$$\frac{24}{16} = \frac{x+11}{x+6}$$

$$\frac{3}{2} = \frac{x+11}{x+6} \\ 3x+18 = 2x+22$$

$$x = 4 \quad \checkmark$$

Reason - 2

b)



Communication - 1

Some responses poorly
set out

students do draw two
separated triangles
generally made errors
some students failed
to state the relevant
test.

generally well done

allowed imperfect diagrams

many students did not
know the symmetry
of odd functions

$$\begin{aligned} \text{c) i) } \alpha + \beta &= -\frac{b}{a} \\ &= -\frac{(-2)}{1} \\ &= 2 \quad \checkmark \end{aligned}$$

$$\begin{aligned} \text{ii) } \alpha\beta &= \frac{c}{a} \\ &= \frac{4}{1} \\ &= 4 \quad \checkmark \end{aligned}$$

$$\begin{aligned} \text{iii) } \alpha^2 + \beta^2 &= (\alpha + \beta)^2 - 2\alpha\beta \quad \checkmark \\ &= (2)^2 - 2 \times 4 \\ &= 4 - 8 \\ &= -4 \quad \checkmark \end{aligned}$$

$$\begin{aligned} \text{iv) } \left(\alpha - \frac{1}{\beta}\right)\left(\beta + \frac{1}{\alpha}\right) \\ &= \alpha\beta + 1 - 1 - \frac{1}{\alpha\beta} \quad \checkmark \\ &= 4 - \frac{1}{4} \\ &= 3\frac{3}{4} \quad \checkmark \end{aligned}$$

Reason - 2

Poor algebra cost marks.

Many students could not arrive at the second line.

Q6 a) i) In $\triangle ABM$ and $\triangle BCN$

$BC = AB$ (sides of a square are equal) $\checkmark$

$BM = CN$ (M & N are midpoints of equal sides)

$\angle ABM = \angle BCN = 90^\circ$ (angles in a square are 90°)

$\therefore \triangle ABM \equiv \triangle BCN$ (SAS) $\checkmark$

Communication - 3

ii) Let $\angle NBC = x$

$\therefore \angle BAM = x$ (corr. angles in cong. triangles) $\checkmark$

$\angle ABN = 90 - x$ ($\angle ABC = 90^\circ$)

$\therefore \angle BPA = 180 - (90 - x) - x$ (angle sum of a triangle is 180°) $\checkmark$

$= 90^\circ$

$\therefore BN \perp AM$

Reason - 2

2 facts correct = 1 mark
3 facts correct = 2 marks
Correct Test = 1 mark

Watch your reasons. Students make up reasons to suit themselves

PLEASE LEARN

Very poorly done.

$$b) y = x^3 - x^2$$

$$\frac{dy}{dx} = 3x^2 - 2x \quad \checkmark$$

$$\therefore 3x^2 - 2x = 8$$

$$3x^2 - 2x - 8 = 0$$

$$(3x+4)(x-2) = 0$$

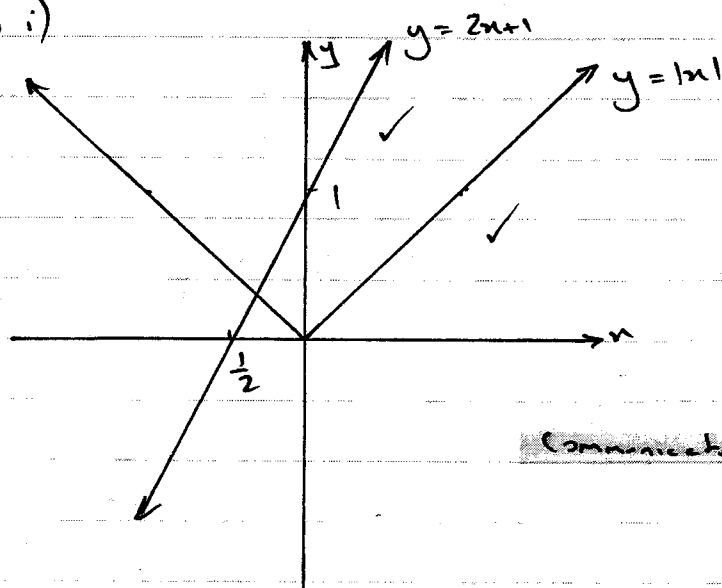
$$x = 2 \quad x = -\frac{4}{3} \quad /$$

Calculus 2

Differentiate was done ok but most students did not know what to do with these.

A lot of students don't know how to factorise.

c) i)



Communication 2

Students need to draw neat graphs labelling all important features.

ii) Any pt of intersection between $y = |x|$ and $y = 2x + 1$ is a solution to $2x + 1 = |x|$

As there is one point of intersection $\therefore$ one solution.

Communication 1

Some students did not identify one point of intersection as one solution

$$iii) 2x + 1 = |x|$$

$$2x + 1 = x \quad \text{or} \quad -2x - 1 = x$$

$$x = -1$$

$$-1 = 3x$$

$$\text{Test } x = -1$$

$$x = -\frac{1}{3} \quad \checkmark$$

$$\text{LHS} = -1$$

$$\text{Test } x = -\frac{1}{3}$$

$$\text{RHS} = 1$$

$$\text{LHS} = \frac{1}{3}$$

$\therefore$ not a solution

$$\text{RHS} = \frac{1}{3}$$

$\therefore x = -\frac{1}{3}$ is only solution. $\checkmark$

A lot of students failed to check their solutions. Even though in part (ii) we know there is only one solution, students still wrote 2 here.

Watch setting out of checking LHS / RHS.

Q7 a) $y = (x+2)^{1/2}$
 $\frac{dy}{dx} = \frac{1}{2}(x+2)^{-1/2} \times 1$
 $= \frac{1}{2\sqrt{x+2}}$ ✓

when $x=7$ $\frac{dy}{dx} = \frac{1}{2\sqrt{7+2}}$
 $= \frac{1}{2 \times 3}$
 $= \frac{1}{6}$

$\therefore m_{\text{tangent}} = \frac{1}{6}$ at P

$\therefore m_{\text{normal}} = -6$ at P ✓ as $\text{tang} \perp \text{normal}$

using $y - y_1 = m(x - x_1)$

$y - 3 = -6(x - 7)$

$y - 3 = -6x + 42$

$6x + y - 45 = 0$ ✓

Calculus-3

ii) $m = \tan \beta$

$\tan \beta = | -6 |$
 $= 6$

$\beta = 81^\circ$

$\theta = 180^\circ - 81^\circ$

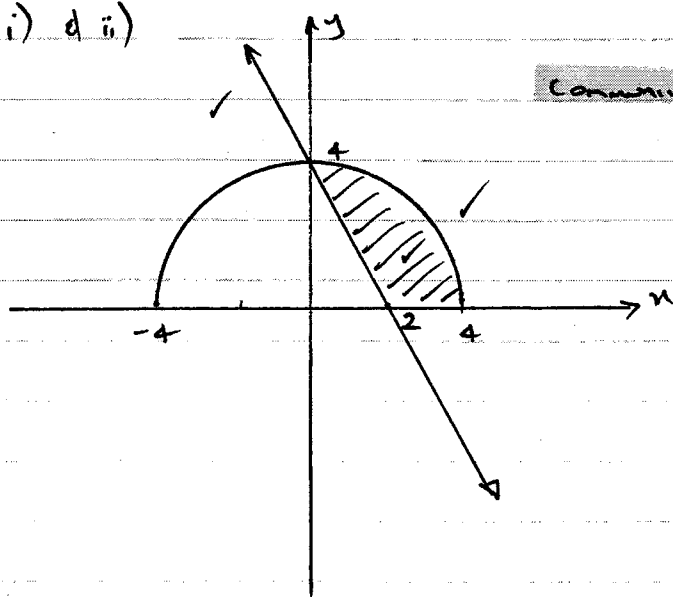
$= 99^\circ$ ✓

- the differentiation was done well but the substitution was done poorly !!

It was a shame that poor algebra skills rather than calculus skills that cost marks.

- don't forget that a negative gradient means an obtuse angle

b) i) & ii)



Construction-3

- the semi-circle was done well but the line was done poorly
 - must have to find x and y-intercepts

$$\text{ii) Area} = \frac{1}{4} \times \pi \times 4^2 - \frac{1}{2} \times 2 \times 4 \\ = (4\pi - 4) \text{ units}^2 \quad \checkmark$$

Reason - 1

- even though I wasn't
sketch an exact answer
is better.

$$\text{c) i) LHS} = \tan \theta + \cot \theta$$

$$= \frac{\sin \theta}{\cos \theta} + \frac{\cos \theta}{\sin \theta}$$

$$= \frac{\sin^2 \theta + \cos^2 \theta}{\sin \theta \cos \theta} \quad \checkmark$$

$$= \frac{1}{\sin \theta \cos \theta}$$

- done well

$$\text{ii) } \frac{1 + \cot \theta}{\cos \theta} - \frac{\sec \theta}{\tan \theta + \cot \theta}$$

$$= \frac{1 + \frac{\cos \theta}{\sin \theta}}{\frac{1}{\sin \theta}} - \frac{1}{\frac{1}{\sin \theta \cos \theta}} \quad \checkmark \text{ from i)}$$

$$= \left(1 + \frac{\cos \theta}{\sin \theta}\right) \times \sin \theta - \frac{1}{\cos \theta} (\sin \theta \cos \theta)$$

$$= \sin \theta + \cos \theta - \sin \theta$$

$$= \cos \theta$$

$$\therefore \cos \theta = -1 \quad \checkmark$$

$$\therefore \theta = 180^\circ \quad \checkmark$$

Reason - 3

- this was a hard
question but it was
algebra skills rather than
trig skills that prevented
getting the question out

- handy hint: although
this doesn't always work
it is a good idea to
change $\tan \theta$, $\cot \theta$ to
 $\frac{\sin \theta}{\cos \theta}$ and $\frac{\cos \theta}{\sin \theta}$

$$\sec \theta = \frac{1}{\cos \theta} \quad \text{and} \quad \csc \theta = \frac{1}{\sin \theta}$$